Appendix: Fine-Grained Ablation Study Details

Table 1: Detailed Ablation Study of Individual Components in K-FedICGL

AL	CF	METR-LA (Speed)			PeMSD4 (Flow)		
		MAE	RMSE	MAPE(%)	MAE	RMSE	MAPE(%)
×	×	3.09	6.14	8.32	23.05	36.12	14.75
✓	×	2.97	5.91	8.02	22.25	34.95	13.92
×	✓	3.02	6.03	8.16	22.68	35.48	14.35
✓	✓	**2.88**	**5.74**	**7.72**	**21.58**	**34.18**	**13.44**

*Note: **AL** denotes the Server-side Invariant Subgraph Alignment (Section 3.3). **CF** denotes the Local Counterfactual Contrastive Optimization (Section 3.4). All variants retain the foundational Knowledge Graph prior.*

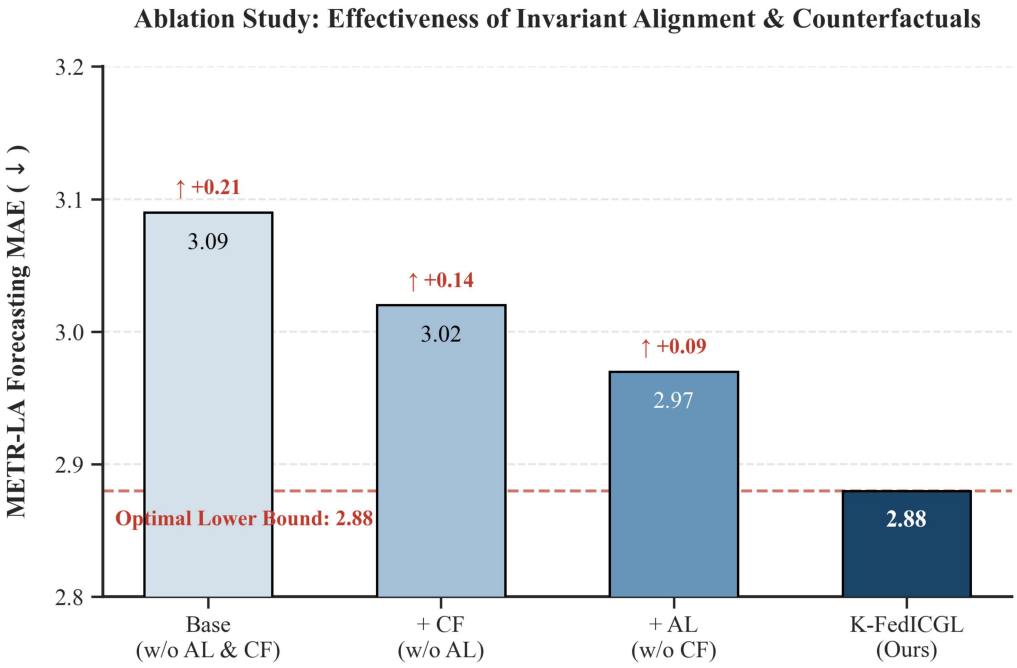

Figure 1: Visual representation of the ablation study on the METR-LA dataset. The red annotations (↑) indicate the ablation penalty (increase in MAE) incurred when removing specific components, highlighting their indispensable roles in achieving the optimal lower bound.